

PLANNING PACK

Phase Plan — With Revisions

Each phase's original goal and cut list, what actually shipped, and the revisions in between — what changed, and what made it change. Pulled from git history and the build's own record of the prompts that changed it.

OC Venue Activation Knowledge Base · August 2026

How to Read This

The individual pivots below were documented as they happened — in git commits, and in a running record of the prompts that changed the architecture — but never pulled into one place until now. This is that document.

A NOTE ON DATING

Git commit timestamps for Phase 1 and Phase 2 are batched on the same day, because of a real process gap covered under Phase 1 below — substantial work happened before anything was committed. Where a specific date matters here, it's the date a decision was made, not the commit date.

Phase 1 — Prove the Pipeline

Goal: one source type ingested end to end — chunk, embed, store — with semantic search returning relevant, cited results. Deployed and demonstrable.

Not in this phase, as planned

Hybrid search, multiple source types, re-ingestion safety, a generation/chat layer, a source-management view.

What shipped

Fullerton's municipal code chapters on special events (15.58) and Conditional Use Permits (15.70), plus the actual fee schedule — 4 documents, 35 chunks, embedded and stored in Supabase with pgvector. Every chunk carries its source section as metadata. Deployed on Railway. Tagged phase-1.

Revisions during this phase

Embedding model: OpenAI → gte-small

The original default was OpenAI's embedding API. Pushing on "why" surfaced a real cost and account-setup burden with no offsetting benefit at this scale — gte-small running inside a Supabase Edge Function is free, needs no separate account, and switching later only means re-embedding existing chunks. Decided before Phase 1 shipped.

Deployment platform: an open Railway-vs-Vercel choice → Railway, confirmed as a hard requirement

Deployment platform had been reasoned from a phase-planning doc, not the actual task brief. Asking directly whether the brief required anything specific sent the process back to read the brief PDF for the first time — it names Railway as a required deliverable three separate times. Without that question, deployment likely would have moved to Vercel: technically "a live app," but not the actual assignment. The single largest correction in the project.

Git discipline: no phase-tagging system → tags as permanent snapshots

By the time Phase 2 was ready to start, a check-in surfaced that despite substantial Phase 1 work, nothing had actually been committed beyond the repo's initial empty commit. This produced the tagging discipline used for the rest of the build (phase-1, phase-2, phase-3 as permanent snapshots) — and, later, the deployment strategy itself: one evolving Railway URL plus git tags as the durable record, over the cost of running a separate live environment per phase.

Demo: the raw hybrid-search page, preserved as the Phase 1–2 view, returning cited passages for a real Fullerton permit question.

Phase 2 — Widen the Intake

Goal: all three required ingestion methods working — an API, a live website, a document upload — with source metadata and citations throughout, and re-ingestion that doesn't create duplicates.

Not in this phase, as planned

Hybrid search tuning, a generation/chat layer, real venue pricing data (known to be open, treated as a Phase 3 concern).

What shipped

50 documents, 89 chunks, three ingestion methods — Google Places API (43 real Fullerton venues/gyms), the OC Fire Authority's special-event permit page (live website), and California's ABC alcohol permit reference forms (document upload). A source-management view. Idempotent re-ingestion via a unique-key constraint and an upsert pattern. Tagged phase-2.

Revisions during this phase

API and live-website sources: Yelp Fusion + Eventbrite → Google Places + OCFA

Both were the original candidates when the source map was first planned. Yelp Fusion turned out not to be free anymore (30-day trial, then \$229+/month) — caught by checking current pricing rather than trusting the assumption. Eventbrite's public search API had been retired. Both swapped for sources that actually worked — and, more durably, this established a pattern for the rest of the build: verify current pricing and terms before committing, rather than trust training data.

Market-rate pricing source: Peerspace → blocked, left as an honest gap

Peerspace, the planned source for real rental pricing, bot-blocked automated access. Rather than force a workaround, this was left as a documented, known gap — the Google Places venue chunks got an explicit "not an actual rental rate" caveat baked into their own text at ingestion time, so the gap carried a correct caveat forward. Closed in Phase 3, not here.

Idempotent ingestion, added mid-phase, reactively

Not originally a distinct line item so much as a lesson already learned by then: a Phase 1 duplicate-document bug (a rejected subagent attempt re-run from scratch, not realizing the first attempt had already succeeded) had already been fixed by hand once. The unique-key constraint and upsert pattern built for Phase 2 exist specifically so that bug can't repeat.

Demo: the source-management view showing all three ingestion methods' output side by side, plus safely re-running ingestion on an already-loaded source with no duplicates appearing.

Phase 3 — Sharpen It

Goal: hybrid search or reranking, a chat interface with grounded and cited answers, a knowledge-graph-or-agentic-retrieval experiment, and evaluation against the question set.

Not in this phase, as planned

A knowledge graph (deliberately not chosen — see below), Phase 4's multi-city expansion, Fullerton insurance/COI data (never published anywhere by the city itself; left out on purpose rather than guessing a generic California number).

What shipped, in the order it actually happened

1. Baseline evaluation first, before touching generation

Rather than building the chat layer and evaluating it after the fact, all 13 test questions were run against Phase 2's raw search first, human-graded (3 GOOD / 5 PARTIAL / 5 FAIL), and written up — a deliberate methodological choice so the generation layer would have a real number to be measured against, not just a "feels better" impression.

2. A grounded, cited chat layer (Google Gemini)

Answers only from retrieved passages, cites every claim, explicitly declines rather than guessing. Caught one real bug: an initial regex-based refusal check false-positively on a genuinely good answer whose text happened to contain a caveat phrase. Fixed by switching to structured JSON output with an explicit has-answer field, instead of parsing prose for signals.

3. Real market-rate data, closing the Phase 2 gap

Direct pushback — "what was the actual purpose of including all those venues?" — surfaced that the revenue-benchmarks table had existed in the data model since Phase 1 planning but was never populated. Seven Fullerton venues' real starting prices were sourced from Zola, each individually address-verified (the initial pull included venues actually in Irvine, Costa Mesa, and Anaheim, since the marketplace's default search radius isn't city-scoped).

4. Reranking + an agentic-retrieval experiment, chosen over a knowledge graph

The fee and venue-pricing tables are already relational — this corpus is fee/permit/pricing facts, not an entity-relationship network a graph would meaningfully improve on. Combined with the brief's own warning that knowledge-graph pipelines "have a special talent for eating weekends," agentic retrieval was the better-fitted experiment, and it directly targeted a diagnosed failure (a gym-permit question whose correct answer existed in the corpus but never matched that phrasing). The two techniques target genuinely different failure modes: reranking promotes a correct-but-buried chunk already retrieved; the agentic requery step rewrites the search itself when nothing retrieved is actually relevant. A first version of the reranker's sufficiency threshold was miscalibrated — too strict, triggering unnecessary second searches and once, a wrongly-conservative refusal on an already-good answer — caught by testing a previously-good question against the new code before trusting it, and fixed before it shipped.

5. Full re-evaluation against the finished chat layer

All 13 questions re-run end to end: **0 FAIL**, down from the baseline's 5 — including eliminating the two most dangerous baseline failures (a different jurisdiction's permit fee presented as Fullerton's own threshold at 0.876 similarity; an unrelated fee chunk presented as an insurance answer).

Tagged phase-3.

Demo: the chat interface — a gym-permit question that visibly triggers the "search refined" indicator and returns a correctly cited answer, next to an out-of-scope question that's honestly declined.

What This Leaves for Phase 4 (Not Started, By Design)

One decision is already banked for when the nine-city Central Orange County expansion starts: **city disambiguation has to be a metadata filter applied before vector ranking, not left to embeddings to disambiguate**. Nine cities' fee ordinances are structurally near-identical — "a special event permit fee of \$X shall be set by resolution," repeated per city — so similarity search alone risks matching the right *kind* of fact to the wrong city. This surfaced from asking "will this be an issue if I scale to Orange County?" early on, well before Phase 4 planning was otherwise underway, and is saved in both this plan and persistent project memory so it isn't lost before that work starts.